

Institute of Actuaries of India

Subject SA4: Pension & Other Employee Benefits

November 2013 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the paper setters with the aim of helping markers of scripts so as to have a framework and be consistent while evaluating answers. The solutions given are only indicative. It is realized that there could be other points as valid answers and the marker may give credit for any such alternative approach or interpretation which the marker considers to be appropriate.

Solution 1 :**i)**

1. Data for actives, member pensioners and spouse beneficiaries comprising of the following information
2. Active data:
 - Employee ID: for identification
 - Designation: to ensure that only eligible members' data has been provided
 - Date of Birth: for current age and the date of retirement for becoming eligible for the benefit
 - Date of Joining: for estimating the past service liability as on valuation date
 - Benefit salary: for estimating the pension and accommodation benefits on retirement in future
 - Spouse Date of Birth: for calculating the joint life pension payable to member and spouse from retirement
3. Pensioner data:
 - Member ID: for identification
 - Member Date of Birth: for calculating the joint life pension payable to member and spouse during retirement
 - Pension amount: for estimating the actuarial liability for retiree pension
 - Spouse Date of Birth: for calculating the joint life pension payable to member and spouse during retirement
 - Accommodation amount: for estimating the actuarial liability for accommodation benefit
4. Beneficiary data:
 - Beneficiary ID: for identification
 - Beneficiary Date of Birth: for calculating the single life pension payable to Beneficiary
 - Pension amount: for estimating the actuarial liability for Beneficiary pension

Given that the individual benefits are material it would be preferable to obtain the spouse information rather than using approximation

(4 Marks)**ii)**

1. The following assumptions are required for the valuation
 - Discount rate
 - Rate of escalation in salary
 - Mortality (pre-retirement)
 - Mortality (post retirement)
 - Attrition rates
 - Rate of Pension increase
 - Early retirement rate
 - Expected rate of return on asset

2. Additional assumptions required for projection

New entrants profile – average age, average benefit salary and percentage of new entrants in future years (maintain population for instance)

The assumptions can be classified into financial and demographic assumptions. The financial assumptions determine the amounts to be paid while the demographic assumptions determine the timing of payment in the future

The assumptions have to be set in accordance with the guidance as per Actuarial Practice Standard 26 and paras 78 -91 of AS 15 (R).

The key criteria to be considered for setting the assumptions are that they have to be set from a long term perspective, using best estimate basis with sensitivity around key assumptions like salary escalation, attrition and mortality post retirement to demonstrate impact due to change in future. Also it would be important to ensure that the different assumptions used are consistent. For instance the rate of pension increase and inflation outlook.

Also it would be important to look at the past scheme experience on account of the assumptions especially salary escalation, expected rate of return on asset and attrition as the experience for mortality may not be sufficient and credible to arrive at a suitable basis.

In addition the following aspects have to be considered while setting the assumptions.

1. Industry benchmark specific to top management, if available
2. Company outlook for the future
3. Current Economic scenario and outlook that would impact the salary escalation and attrition experience
4. Demographic profile for the purpose of new entrants covering average entry age, average monthly salary
5. Inflation outlook and projections
6. Asset profile, income yield, market value, income growth, reinvestment rate and others.

(6 Marks)

iii) The data as at 31 March 2013 is taken as below based on information provided and assumptions as stated (2);

(accounting period assumed as financial year ending 31 March 2013. Any other reasonable assumptions for the accounting period is also acceptable provided it is clearly stated)

Data summary:

No of actives: 28

Average age: 45 (assumed to be same as two older people have retired. Any other logical basis is acceptable too)

Average past service: 5.5 years (assumed to have reduced as two older people have retired. Any other logical basis is acceptable too)

Average monthly benefit salary: INR 185,000* 1.1 = INR 203,500

No of retirees: 10

Average age: 62 (assumed to be same as two new retirees have been added. Any other logical basis is acceptable too)

Average pension benefit: $\text{INR } 65,000 * 1.04 = \text{INR } 67,600$

Average accommodation benefit: $\text{INR } 25,000 * 1.02 = \text{INR } 25,500$ (assumed to have increased due to the retirees this year)

No of beneficiaries: 2 (no change assuming no deaths during the year)

Average age: 59 (increase by one year)

Average pension benefit: $\text{INR } 45,000 * 1.04 = \text{INR } 46,800$

For the purpose of the disclosure, the assumptions at the beginning and end of the year are assumed as below (1)

	31-Mar-12	31-Mar-13
Discount rate	8%	8%
Rate of escalation in salary	9%	9%
Mortality (pre-retirement)	LIC (94 - 96) Ultimate	LIC (94 - 96) Ultimate
Mortality (post retirement)	Annuitants mortality 96-98	Annuitants mortality 96-98
Attrition rates	2% p.a.	2% p.a.
Rate of Pension increase	4% p.a.	4% p.a.
Early retirement	Ignored	Ignored
Expected rate of return on asset	8%	8%
(provide marks for any other reasonable assumption basis)		

For the purpose of the disclosure, the liability and asset position are assumed as below (1);

Balance sheet liability : $\text{INR } 2250 \text{ M} * 1.1 = \text{INR } 2475$ (allowing for increase in salary, benefit amounts and interest cost) [provide marks for any other logical answer]

Value of asset : $\text{INR } 900 \text{ M} * 1.05 = \text{INR } 945$

Benefit payment : $\text{INR } 11 \text{ M}$ (basis calculation as below)

Contribution : $\text{INR } 11 \text{ M}$ (assuming equal to benefit payment)

Average monthly Pension of $\text{INR } 65,000$ and monthly accommodation of $\text{INR } 25,000$ assumed to be paid for the first six months for 8 retirees

Average monthly Pension of 67600 and monthly accommodation of $\text{INR } 25,500$ assumed to be paid for the balance six months for 10 retirees

Considering that pension increase happens in the middle of the year and the two members retired in the middle of the year

Average monthly Pension of $\text{INR } 45,000$ and $\text{INR } 46,800$ assumed to be paid for the first six months and balance six months respectively for 2 beneficiaries

VALUATION AS ON 31 03 2013 (assumed as 31 March 2013)**NAME OF THE PLAN:****Name and address of the Enterprise: ABC Company Limited**

I have been requested by the ABC COMPANY LIMITED to calculate the accounting expenses associated with the DB Pension Plan for Top Management for the financial year ended 31 03 2013 in terms of Accounting Standard (AS) 15 (revised 2005) issued by the Institute of Chartered Accountants of India.

The results set out in this Report and its annexure are based on requirements of AS 15 (revised 2005) and its application to the Plan. They have been prepared for the specific requirements of AS 15 (revised 2005) and should not be used for any other purpose. In particular this Report does not constitute a formal funding actuarial valuation of the Plan and does not present any recommendation of contributions or funding levels.

This Report is provided solely for the Company's use and for the specific purposes indicated above. Except where I expressly agree in writing, it should not be disclosed or provided to any third party, other than as provided below. In the absence of such consent and an express assumption of responsibility, no responsibility whatsoever is accepted by me for any consequences arising from any third party relying on this Report or any advice relating to its contents. The Company may make a copy of this Report available to its auditors, but I make no representation as to the suitability of this Report for any purpose other than that for which it was originally provided and accept no responsibility or liability to the Company's auditors in this regard. The Company should draw the provisions of this paragraph to the attention of its auditors when passing this Report to them.

Summary of Results

2.1 The table below shows a summary of the key results for year ended 31 03 2013.

Assets/Liabilities (INR M)	31 03 2012	31 03 2013
1 Present Value of obligation	(2,250)	(2,475)
2 Fair Value of Plan Assets	900	945
3 Net asset/(liability) recognised in balance sheet	(1,350)	(1,530)

Employer Expense (INR M) Year ending 31 03 2013

1 Current Service Cost	200
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(approximately taken as $2250 \times 1.08/6 \times 0.5$ assuming service cost as opening liability for actives taken as 50% of liability plus interest cost per year of service, where 6 years is the average past service; allow for any other logical basis; reduce 4 marks if values and calculations are not provided)

2 Total Employer expense	191
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1 Current Service Cost represents the cost associated with the current fiscal year benefit accruals. Total Employer expense is the expense under AS15 (Revised, 2005), inclusive of Current Service Cost, Actuarial Gain/loss and net interest.

2 The calculations have been based on the membership information for the Plan as at 31 03 2013 as supplied by the company. A summary of the data is given below.

Summary of Membership Data

	31-Mar-12	31-Mar-13
No of actives:	30	28
Average age:	45	45
Average past service (years)	6	5.5
Average monthly benefit salary:	185,000	203,500
No of retirees:	8	10
Average age: (years)	62	62
Average pension benefit:	65,000	67,600
Average accommodation benefit:	25,000	25,500
No of beneficiaries:	2	2
Average age: (years)	58	59
Average pension benefit:	45,000	46,800

3.2 The company has provided us with the fair value of assets of the Plan as at 31 03 2013. A complete description of the amounts included in the fair value of assets is shown below. We have also been provided with information relating to the contribution and the benefits paid out during the year ended 31 03 2013.

3.3 In preparing this Report we have relied on the completeness and accuracy of the information provided to us orally and in writing by or on behalf of the Company and its advisers. We have not completed any detailed validation checks on the information provided.

We have, however, carried out broad consistency checks.

4 CONCLUSIONS

2 The assumptions and methodology used in compiling this Report are consistent with the requirements of AS15 (Revised 2005).

4.2 The key assumptions used in the calculations are as under.

The principal assumptions are the (1) discount rate & (2) Salary Increase.

The discount rate should be based upon the market yields available on Government bonds at the accounting date with a term that matches that of the liabilities & the salary increase should take account inflation, seniority, promotion and other relevant factors.

The financial assumptions employed for the calculations as at 31 03 2012 and 31 03 2013 are as follows

	31-Mar-12	31-Mar-13
Discount rate	8%	8%
Rate of escalation in salary	9%	9%
Mortality (pre-retirement)	LIC (94 - 96) Ultimate	LIC (94 - 96) Ultimate
Mortality (post retirement)	Annuitants mortality 96-98	Annuitants mortality 96-98
Attrition rates	2% p.a.	2% p.a.
Rate of Pension increase	4% p.a.	4% p.a.

Early retirement	Ignored	Ignored
Expected rate of return on asset	8%	8%

4.3 The results are particularly sensitive to some assumptions, such as the discount rate level of salary inflation, level of assumed price inflation and mortality. A decrease in the discount rate assumed or an increase in salary inflation will lead to an increase in reported cost.

4.4 The methodology used in the calculations is set out below.

We have used the Projected Unit Credit (PUC) actuarial method to assess the Plan's liabilities, including those related to death-in-service and incapacity benefits. Under the PUC method a "projected accrued benefit" is calculated at the beginning of the year and again at the end of the year for each benefit that will accrue for all active members of the Plan. The "projected accrued benefit" is based on the Plan's accrual formula and upon service as of the beginning or end of the year, but using a member's final compensation, projected to the age at which the employee is assumed to leave active service. The Plan Liability is the actuarial present value of the "projected accrued benefits" as of the beginning of the year for active members and pensioners.

4.5 The benefits valued in this Report are summarised below.

Principal Plan Provisions:

We give below a summary of the principal rules of the Plan.

Retirement Age: 65 years

Benefits on retirement:

Pension of 60% of last drawn salary with inflationary adjustments per annum payable to the retiring CEO and his/her spouse for life with 50% reversion to spouse.

A lump sum of 25 lacs at retirement.

Annual benefit of 25% of last drawn salary payable for life of the retiring member to support for accommodation with no increases.

Vesting criteria: None

Benefit on death or disability or resignation: Nil

4.6 We have made full actuarial valuations as at the start and end of the accounting period, based on member data and plan information provided to us at these dates.

4.7 The definitions of various terms used in the Report are given in the GLOSSARY below.

4.8 The full results of our calculations are set out in Disclosure Table in the Annexure to this Report.

4.9 We would be pleased to discuss this Report with you.

ABC COMPANY LIMITED,

I. Assumptions:

	31 03 2012	31 03 2013
Discount Rate	8.00%	8.00%
Rate of increase in Compensation levels	9.00%	9.00%
Rate of Return on Plan Assets	8.00%	8.00%
Expected Average remaining working lives of employees (years)		

II. Table Showing Changes in Present Value of Obligations :

All figures in INR M	31 03 2013
Present Value of Obligation as at the beginning of the year	2,250
Acquisition adjustment	0
Interest Cost	180
Past Service Cost	0
Current Service Cost	200
Curtailment Cost / (Credit) Settlement Cost / (Credit)	0
Benefits paid	(11)
Actuarial (gain)/ loss on obligations	(144)
Present Value of Obligation as at the end of the year	2,475

III Changes in the Fair value of Plan Assets

	31 03 2013
Fair Value of Plan Assets at the beginning of the year	900
Acquisition Adjustments	0
Expected Return on Plan Assets	72
Contributions	11
Benefits Paid	(11)
Actuarial Gain /(loss) on Plan Assets	(27)
Fair Value of Plan Assets at the end of the year	945

IV ACTUARIAL GAIN / LOSS RECOGNIZED

	31 03 2013
Actuarial gain/(loss) for the year - Obligation	144
Actuarial (gain)/loss for the year - Plan Assets	27
Total (gain) / loss for the year	117
Actuarial (gain) / loss recognized in the year	117
Unrecognized actuarial (gains) / losses at the end of year	0

V. THE AMOUNTS TO BE RECOGNIZED IN BALANCE SHEET AND STATEMENTS OF PROFIT AND LOSS

	31 03 2013
Present Value of Obligation as at the end of the year	(2475)
Fair Value of Plan Assets as at the end of the year	945
Funded Status	(1530)
Unrecognized Actuarial (gains) / losses	0

Net Asset / (Liability) Recognized in Balance Sheet (1530)

VI. EXPENSE RECOGNIZED IN THE STATEMENT OF PROFIT AND LOSS

	31 03 2013
Current Service Cost	200
Past Service Cost	0
Interest Cost	180
Expected Return on Plan Assets	(72)
Curtailment Cost / (Credit)	0
Settlement Cost / (Credit)	0
Net actuarial (gain)/ loss recognized in the year	(117)
Expenses Recognized in the statement of Profit & Loss	191

GLOSSARY

All definitions of terms are as per AS 15 (R). The following terms are used with explanations given below:

Actuarial Gain/ Loss From one plan year to the next, if the experience of the plan differs from that or Loss anticipated using the actuarial assumptions, an actuarial gain or loss occurs. For example, an actuarial gain would occur if the plan assets earned 12% for the year while the assumed rate of return used in the valuation was 8%. Other causes of actuarial gains or losses would include changes in actuarial assumptions and / or demographic changes in the population profile.

Balance Sheet asset/ (liability) The sponsor's balance sheet asset/(liability) entry, the net recognised amount, Asset/(Liability) is the sum of the cumulative excess of contributions to the plan over net Annual Expense and other plan-related charges to income due either to business combination or accelerated recognition pursuant to AS15(revised) 2005. The difference between this account and the Funded Status is the unrecognised net loss/ (gain) unvested prior service costs [and net transition obligation.]

Funded Status This is the excess/ (shortfall) of the fair value of plan assets over the Plan Liability.

Plan Liability This quantity is the discounted present value of all benefits attributed by the plan's benefit formula to service rendered prior to the measurement date. It is measured using an assumption as to future pay levels.

Service Cost This is the discounted present value of benefits attributed by the plan's benefit formula to services rendered by employees during the accounting period. It is measured using an assumption as to future pay levels.

Interest Cost The increase in the Plan liability over the accounting period due to interest (the time value of money)

Expected return on plan assets The expected return on plan assets over the accounting period, based on an Return on assumed rate of return.

Net Periodic Benefit Cost This is the profit and loss charge for the accounting period, and comprises the Benefit Cost sum of the service and interest costs less the expected return on assets, plus allowance for amortization of any net liabilities not recognized in the balance sheet.

(15 Marks)

iv) The main differences between AS15 (revised) and FAS87/ 158 are (provide marks for any other reasonable differences)

1. Recognition in Profit & Loss account

- 1.1 The gains and losses arising during the year have to be immediately recognised in the Profit & Loss account under AS15 (revised)
- 1.2 The gains and losses arising during the year are usually amortised over the future working life time under FAS87/ 158. Direct recognition is allowed but very few companies follow that practice under the US GAAP
- 1.3 The unrecognised gains and losses flow into an account called the Accumulated Other Comprehensive Income (AOCI) under US GAAP
- 1.4 While the definition of curtailment and settlements are broadly similar, the treatment of curtailment/ settlement gains and losses are different under the two standards. Under Indian GAAP, the gains and losses are directly recognised in the Profit and Loss account. Whereas under US GAAP, a portion of the unrecognised gains and losses in proportion of the curtailment/ settlement to the overall liability has to be recognised in the Profit and Loss account.

2. Plan design changes

- 2.1 The vested portion of the past service (cost)/ credit arising during the year due to plan design changes have to be recognised immediately under AS15 (revised). The unvested portion will be recognised over the average period to vesting.
- 2.2 The past service (cost)/ credit arising during the year due to plan design changes have to be recognised using a straight line method over the future working life time of the scheme members under US GAAP

3. Discount rate

- 3.1. Government bond yields pertaining to the term of the obligation as on the valuation date is the basis for the discount rate under AS15 (revised)
- 3.2. High quality Corporate bond yields with a rating of AA or above pertaining to the term of the obligation as on the valuation date is the basis for the discount rate under FAS 87/ 158

4. Disclosures

The disclosures under US GAAP require the disclosure of accumulated benefit obligation, expected benefit payments for the future, expense projection, sensitivity to key assumptions etc... which are not required under AS15 (revised) (except the sensitivity to medical inflation under the medical benefit plan)

(5 Marks)

v) Report:**Introduction:**

I have been requested by the management of ABC COMPANY LIMITED to provide the liability for the financial year ended 31 03 2013 and cash flow projections thereafter for a period of five years associated with the Defined Benefit Pension Plan for Top Management. The management has also requested for recommendations on plan design alternatives for the particular plan under consideration.

The results and recommendations set out in this Report and its annexure are based on the framework prescribed under the Actuarial Practice Standard issues by the Institute of Actuaries of India as below;

1. APS 11 : Actuarial Investigation of Retirement Benefit Schemes
2. APS 18 : Retirement Benefit Schemes – Actuarial Reports

The report has been prepared for management information and should not be used for any other purpose. In particular this Report does not constitute a formal funding actuarial valuation of the Plan and does not present any recommendation of contributions or funding levels.

This Report is provided solely for the Company's use and for the specific purposes indicated above. Except where I expressly agree in writing, it should not be disclosed or provided to any third party, other than as provided below. In the absence of such consent and an express assumption of responsibility, no responsibility whatsoever is accepted by me for any consequences arising from any third party relying on this Report or any advice relating to its contents. The Company may make a copy of this Report available to its auditors, but I make no representation as to the suitability of this Report for any purpose other than that for which it was originally provided and accept no responsibility or liability to the Company's auditors in this regard. The Company should draw the provisions of this paragraph to the attention of its auditors when passing this Report to them.

Background and Objective:

1. Purpose of the investigation:

ABC Company Limited is reviewing the Defined Benefit pension scheme currently provided to its top management employees

The intent of the review is to enable

- * Provision of greater flexibility for saving and utilisation to the members of the scheme
- * Decision making by the members rather than benefit as defined in the policy

* Optimising cost and cash flows to the Company

2. Scheme being analysed:

Summary of my understanding of the benefits applying to the members, the data and assumptions used is provided further in this report

Plan provisions:

Retirement Age: 65 years

Benefits on retirement:

Pension of 60% of last drawn salary with inflationary adjustments per annum payable to the retiring CEO and his/her spouse for life with 50% reversion to spouse.

A lump sum of 25 lacs at retirement.

Annual benefit of 25% of last drawn salary payable for life of the retiring member to support for accommodation with no increases.

Vesting criteria: None

Benefit on death or disability or resignation: Nil

Alternative plan design being reviewed has been defined as below:

1. Immediate pay out of the lump sum equivalent of the benefit
2. Pay out of the lump sum equivalent for the pension and accommodation benefit at retirement
3. Pay out of the lump sum equivalent in two equal installments one at age 55 and another at age 60

3. Data used:

	31-Mar-13
No of actives:	28
Average age:	45
Average past service (years)	5.5
Average monthly benefit salary:	203,500
No of retirees:	10
Average age: (years)	62
Average pension benefit:	67,600
Average accommodation benefit:	25,500
No of beneficiaries:	2
Average age: (years)	59
Average pension benefit:	46,800

I have been provided with the data as at 31 March 2013. In preparing this report, I have relied on the completeness and accuracy of the information provided to me orally and in writing by the

Company. I have not completed any detailed validation checks on the data provided. However I have reviewed the data for its reliability

4. Assumptions used:

	31-Mar-13
Discount rate	8%
Rate of escalation in salary	9%
Mortality (pre-retirement)	LIC (94 - 96) Ultimate
Mortality (post retirement)	Annuity mortality 96-98
Attrition rates	2% p.a.
Rate of Pension increase	4% p.a.
Early retirement	Ignored
Expected rate of return on asset	8%

In order to estimate the liability and cash flow projections certain best estimate assumptions have been made. A summary of the same is as provided above. The results of the valuation are highly sensitive to the assumptions used especially the discount rate, salary escalation rate, attrition and post retirement mortality.

Assumptions have been chosen on a best estimate basis and are consistent with the accounting valuation done for ABC Company Limited.

To demonstrate the impact of changes in assumptions on the liability and the cash flows, sensitivity of the numbers to the key assumptions have been calculated as indicated below and presented in the report for management information.

Discount rate sensitivity	+/- 50 basis points
Rate of escalation in salary	+/- 50 basis points
Mortality (post retirement)	Longevity improvement of 3 years
Attrition rates	+/- 200 basis points
Rate of Pension increase	+/- 100 basis points
Expected rate of return on asset	+/- 50 basis points

5. Methodology used

I have used the Projected Unit Credit (PUC) actuarial method to assess the Plan's liabilities, including those related to death-in-service and incapacity benefits

Under the PUC method a "projected accrued benefit" is calculated at the beginning of the year and again at the end of the year for each benefit that will accrue for all active members of the Plan. The "projected accrued benefit" is based on the Plan's accrual formula and upon service as of the beginning or end of the year, but using a member's final compensation, projected to the age at which the employee is assumed to leave active service. The Plan liability is the actuarial present value of the "projected accrued benefits" as of the beginning of the year for active members

In estimating the payments, salaries are projected into the future to the likely time of exit. Assumptions are made about the likelihood of a benefit becoming payable at any future date (the demographic assumptions) and about members' salary increases (the financial assumptions)

The liability and cash flow projections:

Based on the data, assumptions, sensitivity and plan provisions used please find below the liability for the current plan.

Net Liability as at 31 March 2013 (INR M)	
Best estimate basis	1530
Sensitivity	
DR : +50 bp	XX
DR : -50 bp	XX
SE : +50 bp	XX
SE : -50 bp	XX
Mortality post retirement	
(improvement by 3 years)	XX
Attrition : +200 bp	XX
Attrition : - 200 bp	XX
Pension Increase: +50 bp	XX
Pension Increase: -50 bp	XX

Also the cash flow projections for the next five years based on the data, assumptions and plan provisions used are as below.

ending	year	cash flow projected
2014		XX
2015		XX
2016		XX
2017		XX
2018		XX

Draw out some key findings from the sensitivity for information. For instance, a 50 basis point change in Discount rate results in a _% change in liability as at 31 March 2013

Impact of the three suggested design alternatives & accounting/ cash flow impact:

As requested by the Company, I have carried out the valuation under the three alternative plan design scenarios being reviewed as defined below:

1. Immediate pay out of the lump sum equivalent of the benefit
2. Pay out of the lump sum equivalent for the pension and accommodation benefit at retirement
3. Pay out of the lump sum equivalent in two equal installments one at age 55 and another at age 60

I have evaluated the cash lump sum under option one using a discontinuance basis assuming no further accrual of benefits. However the following needs to be taken into consideration when determining discontinuance terms:

- Rights of beneficiaries: The benefits that will be paid to the members of a discontinued defined benefit scheme depend on the rights of the members under the Pension scheme Trust Deed and Rules and overriding legislation. Commonly their rights are the members' accrued benefits (excluding any discretions that may be granted in future), with active members treated as though they had left pensionable service the day before the termination.
- Deferred benefits would be available even for members who have not completed any vesting period. In this situation deferred benefits would be calculated in the same way as for members who have completed the vesting period.
- Other benefit expectations
- the availability and selection of a method of provision of discontinuance benefits and
- The level of available assets. The company should provide for the following benefits in the indicated order of priority if the level of available assets is a determinant to the cash payouts.
 - The cost of administering the cash pay outs
 - The existing pensioners and beneficiaries

- Accrued benefits of existing members
- Future accrual of benefits for existing members and
- Discretionary pension increases in the future

For option two, I have evaluated the lump sum as the cash value required to purchase the benefit from an insurance company at retirement

For option three, I have estimated the cash value equivalent at ages 55 and 60 of the benefit payable at retirement.

The calculations have been made on the basis of the data and best estimate assumptions as described in this report. I have not allowed for any discretionary pension increases in the future for the purpose of these estimates.

The company must consider the following aspects for the cash out

1. Documentation of the change in Trust deed and rules
2. Communication to members
3. Review of the investment strategy in the light of the cash out option chosen
4. Contribution to the fund if the payout is more than the value of the asset
5. Accounting and cash flow impact
6. Settlement of the benefits and
7. Winding up of the trust once all cash payout have been made

Other design considerations

The following factors need to be considered in evaluating other alternatives for the redesign of the scheme taking into account the objectives of the management as described earlier.

- Cost
- Sustainability
- Employee demography and ability to contribute in the event of cash out
- Employee expectation and buy-in
- Legal compliance
- Tax implication
- Market benchmarking

- Winner loser analysis for design features proposed
- Communication requirements
- Administration requirements
- Management buy in
- Availability of suitable options for buy out

(20 Marks)

[Total Marks-50]

Solution 2 : (i)**a)**

Main advantages of funding from the point of view of the employee members of the approved superannuation scheme:

- Funding gives employees better security of benefits as the funds are held with Trustee managed trust and no part of such fund can be reverted back to the employer, such security however is limited by the adequacy of funding.
- If there is a fund built up, adverse developments are less likely to mean that the benefits earned up to a particular time are not provided as expected.
- Funding can protect accrued pensions from an employer going bankrupt, or the employer being unable or unwilling to make future funding payments.
- As Trustees of the fund have fiduciary responsibility and are legally ring-fenced by legal obligations to comply with Indian Trust Act, the degree of security of benefits is high.

(4 Marks)

b)

The different ways in which benefits could be funded are:

- Lump sum in advance
- Regular Funding
- Terminal Funding
- Pay As You Go (PAYG)

Funding Type	Security	Tax Concessions	Cash Flow Planning
Lump sum in advance	? Security depends on the investment vehicle and the risks involved	x Tax concession may not be applicable to all the contribution	? Cashflow required depends on the number and profile of new entrants
Regular Funding	✓ Security depends on the investment vehicle, timing of the funding and the risks involved	✓ Tax concession may be fully applicable to all the contribution	✓ Cashflow required would depend on the chosen method of funding

Terminal Funding	✓ Security depends on the chosen method of benefit payment	x Tax concession may not be fully applicable to all the benefits paid	x Terminal Funding may involve future uncertain cash flows
Pay As You Go (PAYG)	✓	x Tax concession may not be fully applicable to all the benefits paid	x PAYG may involve future uncertain cash flows

✓ = satisfies criteria

? = partially satisfies criteria

x = does not meet criteria

(5 Marks)

c)

Minimum Required Contribution

The term “minimum required contribution” means, with respect to any plan year of a defined benefit plan which is not a multi-employer plan:

(1) in any case in which the value of plan assets of the plan is less than the funding target of the plan for the plan year, the sum of—

- (A) the target normal cost of the plan for the plan year,
- (B) the shortfall amortization charge (if any) for the plan for the plan year, and
- (C) the waiver amortization charge (if any) for the plan for the plan year

(2) in any case in which the value of plan assets of the plan equals or exceeds the funding target of the plan for the plan year, the target normal cost of the plan for the plan year reduced (but not below zero) by such excess.

Target Normal Cost

The term “target normal cost” means, for any plan year, the excess of:

(A) the sum of

- (i) the present value of all benefits which are expected to accrue or to be earned under the plan during the plan year, plus
- (ii) the amount of plan-related expenses expected to be paid from plan assets during the plan year, over

(B) the amount of mandatory employee contributions expected to be made during the plan year.

For purposes of this subsection, if any benefit attributable to services performed in a preceding plan year is increased by reason of any increase in compensation during the current plan year, the increase in such benefit shall be treated as having accrued during the current plan year.

(5 Marks)

(ii)

a) The choice of assumptions used in an actuarial valuation of a pension plan covering aspects from APS 26

Background The actuarial assumptions used in a pension plan valuation are discount rate, withdrawal rate, salary escalation rate, pension escalation rate, mortality (both in service and post retirement) and investments return.

AS 15 (revised, 2005) has set out in paragraphs 78 to 91 the basis upon which the various assumptions to be used in the actuarial valuation for accounting purposes are to be determined.

The assumptions chosen for bank accounting purposes and reporting in the financial statements have to reflect the market conditions as at the Balance sheet date.

APS 26 states that the actuarial assumptions are the enterprise's best estimates of the different variables and the responsibility for setting these financial assumptions rests with the enterprise.

APS 26 further states that the actuary shall advise the client on the interpretation and determination of the various assumptions and comment on the suitability of the assumptions chosen by the client.

Discount Rate

Paragraph 78 of AS 15 (revised, 2005) states that the rate used to discount post-employment benefit obligations should be determined by reference to market yields at the balance sheet date on government bonds. The currency and term of the government bonds should be consistent with the currency and estimated term of the post-employment benefit obligations, in this case the pension benefits. The bank should use the appropriate discount rate based on this prescribed approach.

Salary Escalation Rate

In projecting the salary increases, the bank needs to keep in mind two factors - one is inflation level leading to a general change in salary level. The other is career progression of the bank employees. The bank needs to combine the two and arrive at suitable salary growth levels.

Pension Escalation Rate

In projecting the pension increases, the bank needs to keep in mind two factors – one is the inflation level and the second is the long term nature of the benefit payout. The bank needs to appropriately arrive at a rate combining the two factors.

Attrition Rate

Withdrawal rates during early durations of service and near retirement date will have a significant impact on the actuarial estimates. The future experience will also depend on general economic conditions as also the particular conditions affecting the bank employee population. Further the rates could differ from bank. Hence, the bank needs to carefully choose the assumption which will reflect the future employee experience on a realistic basis.

Mortality

The Indian Assured Lives Mortality (2006-08) Table is the latest published mortality table available on Indian Assured Lives. This could be used for mortality in service. The LIC (1996-98) Annuitants' Mortality Table could be used for mortality post retirement. However, such usage should reflect the actual experience of each Bank vis-à-vis these published Table and consequential modification, if found necessary. The Actuary should be able to carry out such an exercise and make suitable recommendations.

Return on Plan Assets

The expected return on Plan Assets is a component of expense recognized in the statement of Profit and Loss. The difference between expected and actual return on Plan Assets is actuarial gain or loss. The expected return on the Plan Assets has to be based on Market Expectations. at the beginning of the period, for returns over entire life of the related obligations. The Actuary should be able to assist the enterprise as to how best to articulate these market expectations and determine the expected return.

(8 Marks)

b)

The principles to be covered in the IBA Guidelines

1. **The context:** The objective of the IBA guideline is to bring out uniformity in respect of actuarial assumptions, presumably both for funding (to determine contribution rate and adequacy of funding) and expensing (under AS 15 (rev 2005)). The guidelines should cover the principles underlying the actuarial assumptions in respect of these two objective which in the nature of things have to be entity specific. The requirement of 'uniformity' should therefore focus on the uniformity of principles and methodology and certainly not on the results. The principles underlying the determination of the assumptions for expensing have to have roots in what is stated in the AS 15 (rev. 2005) and corresponding Actuarial Practice Standard (APS 26) of the Institute of Actuaries of India. Keeping in view the fact that funding of employee benefits is not regulated and thus there are no regulations around the same, the requirements of Income Tax Authorities do put certain upper limits on the contribution for tax approval purposes. While formulating the intended Guidelines this aspect as well as targeted funding ratio and contribution rates have to be taken in to account.

2. **Bank identity:** As each Bank is a legal entity, hence an 'enterprise' within the meaning of AS 15(rev. 2005) and other relevant Laws such as In come Tax and Companies Act, therefore such guidelines need to reflect each bank's experience and hence have to be principles based encompassing all the Banks.
3. **Key points in the Guidelines:** The Guidelines have to be drawn embedding the following at the least;
 - i) Specific to expensing and reporting as a requirement under AS 15 (rev. 2005)
 - ii) Specific to funding for ensuring adequacy of funds to ensure availability of resources to meet liabilities and rate of contribution including treatment of tax matters as required under Income Tax rules.
 - iii) Creation of Trusts and selection of Trustees so as to ensure that all the IBA Member Banks set up Trusts, select Trustees on similar patter and set requirements on Trustees so as to ensure separation of Trustee functions from Management functions.
 - iv) Investment of Trust fund including matters such as whether funds should be self-administered and/or Insurers to be appointed as Fund Managers with key principles to be followed.
 - v) The guideline could make it clear that the banks should not choose or modify assumptions in a way which can aim to target certain amount of profit/loss.
 - vi) The level of accuracy required and the potential financial significance of errors..
 - vii) Guidelines for appointment of Actuary: what experience, Fellow member of IAI, Professional compliance and accountability.

(10 Marks)

c)

The points that could be covered in the policy to be approved by the board

1. **Bank Specific Policy:** The IBA guidelines require its Bank members to have a policy framework with the objective of implementing the IBA Guidelines. Such a Policy framework therefore have to be specific to each of the Guideline lines and state as to how these will be implemented. The Guidelines being non-Bank specific, the Policy framework should also have experience of the Bank integrated in to it.
2. **The Policy Framework/Document should at the minimum contain the following :-**
 - i) Matters related to Expensing/reporting under AS 15 (rev. 2005) and Funding separately.
 - ii) The method of determination of major assumptions – both demographic and financial, in consultation with an actuary.
 - iii) Appointment of the Actuary (requirements on Actuary with regard to his/her experience could be different for each Bank), experience (whether have worked as Actuary in the Banking industry), whether have been working for the same

Bank and for how many years (Bank may decide to fix maximum tenure for any one Actuary at a time). Also whether the same Actuary should be appointed for AS 15(rev. 2005) report and Funding Actuarial report.

- iv) Appointment of Trustees; how to select and from where including need for independent Trustees.
- v) Mechanism to monitor performance of Trustee functions including the manner in which issues of conflict of interest would be dealt with.
- vi) Rules for appointment of Fund Managers including criteria of selection and performance measurement.
- vii) The Investment Policy including Asset Liability Management framework that will be required of Trustees to adopt and implement and frequency at which such a Policy will be reviewed.
- viii) The manner in which the Trust Fund Accounts will be prepared and presented and Audited including whether Bank Auditor and Trust Fund Auditor could be same..
- ix) The target level of funding including contribution rates and the manner in which deficit/surplus in the fund would be dealt with.

(10 Marks)

(iii)

The RBI Guidelines are expected to aim at Trustees with regard to management of Trust Funds particularly options of Self-administered funds as against funds managed by Insurers. The RBI being regulator of the Banking Industry has to view its role in this regard from the point of jurisdiction and whether objectives such as this should better be achieved through IBA. The Investment of Trust Funds is responsibility of Trustees who have accountability under Indian Trust Act, such Trustees are appointed by each Bank within the framework of its own Board guidelines which may include agreement with the Staff Unions. However, irrespective of the fact whether the stated RBI objectives are achieved by direct (to Banks) or through IBA, the following key points need to be captured in the Guidelines ;

1. The Trust Fund objectives of maximizing investment return within the boundaries of asset safety, availability of cash flow when needed to discharge member benefits and therefore ALM issues. This may differ in matters of details from Bank to Bank.
2. Tax efficiency of the assets in the Trust fund whether invested as self-administered fund or by appointing life insurer/s a Fund Managers.
3. Suitability of the Life Insurer products for such approved superannuation funds; all life insurers may not have slew of same products and all products may not match the ALM requirements of the Trust Funds.

4. Desirability of having add-on schemes such as Group Term Insurance which many of the Life Insurers sell along with the Fund Products and costs associated with the same.
5. Safety of the Funds under Life Insurer management from the perspective of Life Insurer Solvency: Capital, Solvency Margin and similar other parameters.
6. How the annuity purchase when pensions fall due to be started will be arranged, whether the Trust Fund is allowed to pay the pension/annuity from the fund and would then opt for it and/or whether annuities would be purchased from life insurers at a competitive rate.
7. IRDA regulatory oversight on Life Insurers in respect of schemes of Fund Management and how best such oversight would better the security of the members as against self-administered funds which are driven by requirement of Tax Authorities and oversight from EPFO/MOF.

(8 Marks)

[Total Marks-50]
